

Nate Woodward

LC T P Spring symposium: Theoretical Physics and AI

Reasoning-Fine-Tuning: SFT

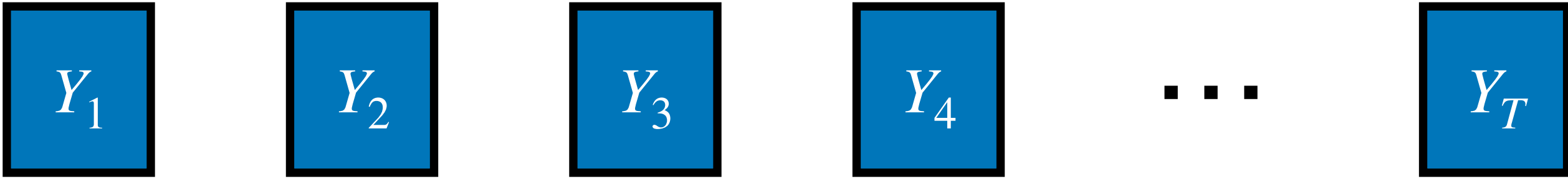


Problem Prompt (X)

$$-\log P(Y|X) = -\sum_{t=1}^T \log P(Y_T|X, Y_{<t})$$

Teacher Model Solution

Teacher Solution per token



Student Probability per token

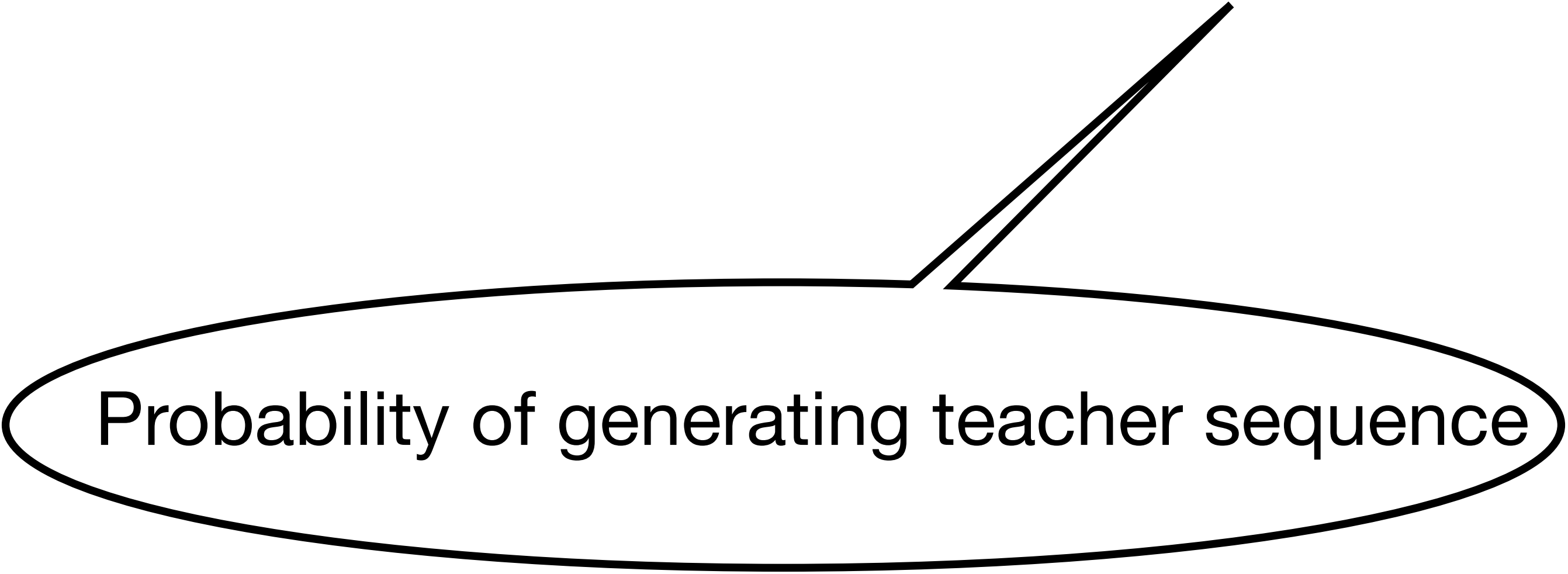
$$P(Y_1 | X)$$

$$P(Y_2 | X, Y_1)$$

...

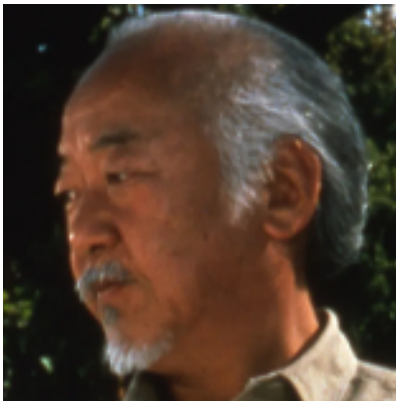
$$P(Y_T | X, Y_{<T})$$

Loss



Probability of generating teacher sequence



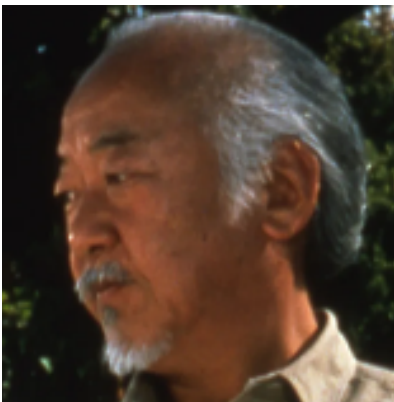


Reasoning Fine-Tuning: SFT

Problem Prompt (X)

Teacher Model Solution

Teacher Solution per token



Y_1

Y_2

Y_3

Y_4

...

Y_T

Loss

$$-\log P(Y|X) = -\sum_{t=1}^T \log P(Y_T|X, Y_{<t})$$

Student Probability per token



$P(Y_1|X)$

$P(Y_2|X, Y_1)$

...

$P(Y_T|X, Y_{<T})$

Probability of generating teacher sequence

Our Approach